

APM1110_FA5_Paglinawan_Zeus

Zeus Paglinawan

2026-09-18

ITEM 1 - DISCRETE RANDOM VARIABLES

Item 1(a) - Find k

Given:

$$P(X = x) = k(2x + 1), \quad x = 1, 2, 3, 4$$

$$\sum P(X = x) = 1$$

(substitute)

$$k(2(1) + 1) + k(2(2) + 1) + k(2(3) + 1) + k(2(4) + 1) = 1$$

$$3k + 5k + 7k + 9k = 1$$

$$24k = 1$$

$$k = \frac{1}{24}$$

$$k = 0.0417$$

```
x <- 1:4
k <- 1 / sum(2*x + 1)
round(k,4)
```

```
## [1] 0.0417
```

Therefore,

$$k = 0.0417$$

ITEM 1(b) - Expected Value and Variance

$$(k = \frac{1}{24})$$

$$P(X = 1) = \frac{3}{24}, \quad P(X = 2) = \frac{5}{24}, \quad P(X = 3) = \frac{7}{24}, \quad P(X = 4) = \frac{9}{24}$$

(formula)

$$E[X] = \sum xP(X = x)$$

(substitute)

$$\begin{aligned}E[X] &= 1 \left(\frac{3}{24} \right) + 2 \left(\frac{5}{24} \right) + 3 \left(\frac{7}{24} \right) + 4 \left(\frac{9}{24} \right) \\E[X] &= \frac{3 + 10 + 21 + 36}{24} \\E[X] &= \frac{70}{24} = 2.9167\end{aligned}$$

Variance:

find $E[X^2]$:

$$\begin{aligned}E[X^2] &= \sum x^2 P(X = x) \\E[X^2] &= 1^2 \left(\frac{3}{24} \right) + 2^2 \left(\frac{5}{24} \right) + 3^2 \left(\frac{7}{24} \right) + 4^2 \left(\frac{9}{24} \right) \\E[X^2] &= \frac{3 + 20 + 63 + 144}{24} \\E[X^2] &= \frac{230}{24} = 9.5833\end{aligned}$$

(formula)

$$\begin{aligned}\text{Var}(X) &= E[X^2] - [E(X)]^2 \\ \text{Var}(X) &= 9.5833 - (2.9167)^2 \\ \text{Var}(X) &= 1.0764\end{aligned}$$

```
p <- k * (2*x + 1)
EX <- sum(x * p)
EX2 <- sum(x^2 * p)
variance <- EX2 - EX^2
round(EX,4)
```

```
## [1] 2.9167
```

```
round(EX2,4)
```

```
## [1] 9.5833
```

```
round(variance,4)
```

```
## [1] 1.0764
```

ITEM 1(c)

$$P(X \geq 3)$$

(values greater than or equal to 3)

$$P(X \geq 3) = P(X = 3) + P(X = 4)$$

(substitute)

$$P(X \geq 3) = \frac{7}{24} + \frac{9}{24}$$

$$P(X \geq 3) = \frac{16}{24}$$

$$P(X \geq 3) = 0.6667$$

```
P_X_ge_3 <- sum(p[x >= 3])
```

```
round(P_X_ge_3, 4)
```

```
## [1] 0.6667
```

ITEM 2 - BINOMIAL DISTRIBUTION

Given:

n = 20

p = 0.08

ITEM 2(a) - Exactly 2 Malicious Requests

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

$$P(X = 2) = \binom{20}{2} (0.08)^2 (0.92)^{18}$$

$$\binom{20}{2} = 190$$

$$P(X = 2) = 190(0.08)^2(0.92)^{18}$$

$$P(X = 2) = 0.2711$$

```
n <- 20
```

```
p_binom <- 0.08
```

```
P_X_2 <- dbinom(2, size = n, prob = p_binom)
```

```
round(P_X_2, 4)
```

```
## [1] 0.2711
```

ITEM 2(b) - At Least 3 Malicious Requests(greater than or equal to 3)

$$P(X \geq 3)$$

$$P(X \geq 3) = 1 - P(X \leq 2)$$

$X \leq 2$ means $X = 0, 1, 2$:

$$P(X \geq 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

(binomial formula)

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

$$1 - p = 1 - 0.08 = 0.92$$

$$P(X = 0)$$

$$P(X = 0) = \binom{20}{0} (0.08)^0 (0.92)^{20}$$

$$P(X = 0) = 0.1887$$

$$P(X = 1)$$

$$P(X = 1) = \binom{20}{1} (0.08)^1 (0.92)^{19}$$

$$P(X = 1) = 0.3282$$

$$P(X = 2)$$

$$P(X = 2) = \binom{20}{2} (0.08)^2 (0.92)^{18}$$

$$P(X = 2) = 0.2711$$

Using the unrounded values:

$$P(X \geq 3) = 1 - P(X \leq 2)$$

$$P(X \geq 3) = 0.2120538$$

$$P(X \geq 3) = 0.2121$$

```
P_X_ge_3_binom <- 1 - pbinom(2, size = n, prob = p_binom)
round(P_X_ge_3_binom, 4)
```

```
## [1] 0.2121
```

ITEM 2(c) - Mean and Standard Deviation

(mean)

$$\mu = np$$

(substitute)

$$\mu = (20)(0.08)$$

$$\mu = 1.6000$$

(standard deviation)

$$\sigma = \sqrt{np(1 - p)}$$

(substitute)

$$\sigma = \sqrt{(20)(0.08)(0.92)}$$

$$\sigma = \sqrt{1.472}$$

$$\sigma = 1.2133$$

```
mean_X <- n * p_binom

sd_X <- sqrt(n * p_binom * (1 - p_binom))

format(round(mean_X, 4), nsmall = 4)
```

```
## [1] "1.6000"
```

```
format(round(sd_X, 4), nsmall = 4)
```

```
## [1] "1.2133"
```

ITEM 3 - POISSON DISTRIBUTION

Given: Average = 4.5 hits per minute

ITEM 3(a) - Exactly 6 Hits

Given:

$\lambda = 4.5$ hits per minute

(formula)

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

(6 hits)

$$P(X = 6)$$

$$P(X = 6) = \frac{e^{-4.5} (4.5)^6}{6!}$$

$$P(X = 6) = 0.1281201$$

Rounded to four decimal places:

$$P(X = 6) = 0.1281$$

```
lambda <- 4.5

P_X_6 <- dpois(6, lambda = lambda)

round(P_X_6, 4)
```

```
## [1] 0.1281
```

ITEM 3(b) - At Most 2 Hits(less than or equal to 2)

$$P(X \leq 2)$$

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

(formula)

$$P(X \leq 2) = e^{-4.5} \left(\frac{4.5^0}{0!} + \frac{4.5^1}{1!} + \frac{4.5^2}{2!} \right)$$
$$P(X \leq 2) = 0.1735781$$
$$P(X \leq 2) = 0.1736$$

```
P_X_at_most_2 <- ppois(2, lambda = lambda)
round(P_X_at_most_2, 4)
```

```
## [1] 0.1736
```

ITEM 3(c) - More Than 10 Hits in 2 Minutes

(2 minutes)

$$\lambda = (4.5)(2)$$
$$\lambda = 9$$
$$P(X > 10)$$
$$P(X > 10) = 1 - P(X \leq 10)$$
$$P(X > 10) = 1 - \sum_{x=0}^{10} \frac{e^{-9} 9^x}{x!}$$
$$P(X > 10) = 0.2940117$$
$$P(X > 10) = 0.2940$$

```
lambda_2min <- 4.5 * 2
P_X_more_10 <- 1 - ppois(10, lambda = lambda_2min)
format(round(P_X_more_10, 4), nsmall = 4)
```

```
## [1] "0.2940"
```

ITEM 4 - CONTINUOUS PROBABILITY & NORMAL DISTRIBUTION

Given:

Mean = 120 milliseconds

Standard deviation = 15 milliseconds

ITEM 4(a) - Longer Than 135 ms

Given:

$$\mu = 120 \text{ ms}$$

$$\sigma = 15 \text{ ms}$$

(formula)

$$z = \frac{x - \mu}{\sigma}$$

(longer than 135ms)

$$P(X > 135)$$

(z-score)

$$z = \frac{135 - 120}{15}$$
$$z = 1$$

$$P(X > 135) = P(Z > 1)$$

(standard normal distribution)

$$P(Z > 1) = 1 - 0.8413$$

$$P(Z > 1) = 0.1587$$

```
mu <- 120
sigma <- 15
P_X_more_135 <- 1 - pnorm(135, mean = mu, sd = sigma)
round(P_X_more_135,4)
```

```
## [1] 0.1587
```

ITEM 4(b) - Between 100 and 130 ms

$$P(100 < X < 130)$$

(two z-scores)

$$z_1 = \frac{100 - 120}{15}$$
$$z_1 = -1.3333$$
$$z_2 = \frac{130 - 120}{15}$$
$$z_2 = 0.6667$$

$$P(100 < X < 130) = P(-1.3333 < Z < 0.6667)$$

(normal distribution)

$$P(100 < X < 130) = P(Z < 0.6667) - P(Z < -1.3333)$$

$$P(100 < X < 130) = 0.6562962$$

$$P(100 < X < 130) = 0.6563$$

```
P_between <- pnorm(130, mean = mu, sd = sigma) -
  pnorm(100, mean = mu, sd = sigma)
round(P_between,4)
```

```
## [1] 0.6563
```

ITEM 4(c) - 95th Percentile

$$P(X \leq x) = 0.95$$

(standard normal value)

$$z_{0.95} = 1.6448536$$

$$x = \mu + z\sigma$$

(substitute)

$$x = 120 + (1.6448536)(15)$$

$$x = 144.6728$$

$$95th\ percentile = 144.6728\ ms$$

```
P95 <- qnorm(0.95, mean = mu, sd = sigma)
round(P95,4)
```

```
## [1] 144.6728
```